Eur. Phys. J. Spec. Top. (2025) 234:4483–4496
https://doi.org/10.1140/epjs/s11734-025-01781-y

THE EUROPEAN PHYSICAL JOURNAL SPECIAL TOPICS

Regular Article

ECG-ResViT: a hybrid CNN-ViT model for efficient ECG signal classification

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Received 24 April 2025 / Accepted 27 June 2025 / Published online 10 July 2025
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Abstract Electrocardiogram (ECG) analysis is a critical tool in the early detection of life-threatening cardiac conditions, such as Arrhythmia (ARR), Congestive Heart Failure (CHF), and Normal Sinus Rhythm (NSR). While deep learning (DL) models, especially Convolutional Neural Networks (CNNs), have demonstrated success in identifying local patterns in ECG signals, their limited capacity to capture long-range dependencies often impacts performance. On the other hand, Vision Transformers (ViTs), known for their strength in modeling global relationships, are underexplored in the context of 1D biomedical signals. This study introduces ECG-ResViT, a novel hybrid DL architecture that combines the local feature extraction power of CNNs—enhanced with residual blocks and dilated convolutions—with the global attention modeling of ViTs. The architecture leverages positional encoding and multi-head self-attention to effectively capture temporal dependencies in ECG signals. Evaluated on benchmark datasets comprising ARR, CHF, and NSR classes, the proposed model achieves an accuracy of 99.13% and ROC-AUC of 99.94%, outperforming several existing state-of-the-art methods. By balancing accuracy, interpretability, and computational efficiency, ECG-ResViT demonstrates strong potential for real-time clinical deployment and highlights the value of hybrid approaches in advancing biomedical signal classification.

1 Introduction

Cardiovascular diseases (CVDs) remain the leading cause of death globally, claiming an estimated 17.9 million lives each year, which accounts for nearly 32% of all global deaths as reported by World Health Organization (WHO) [1]. Among these, timely diagnosis and classification of heart conditions like ARR, CHF, and NSR are critical in preventing severe complications. ECG signal analysis has proven to be a fundamental tool in clinical settings for monitoring heart activity and detecting abnormalities.

While experienced medical professionals are proficient at interpreting ECG signals, manual analysis can be time-consuming, subjective, and susceptible to variations, especially in emergency or real-time settings. This is where automated DL-based solutions come into play—offering speed, consistency, and the potential for real-time clinical assistance.

In recent years, DL models have gained popularity for ECG classification due to their ability to automatically learn complex patterns from raw data. CNN, in particular, are effective in detecting localized features like P waves, QRS complexes, and T waves. However, a known limitation of CNNs is their difficulty in modeling long-range temporal dependencies, which are equally important in analyzing the full scope of cardiac activity. To address this, enhancements, such as residual connections and dilated convolutions, are integrated into the CNN blocks. These techniques help the model retain important gradients during training and allow a wider receptive field, respectively—without increasing the computational load significantly.

On the other hand, Transformer-based models—originally developed for natural language tasks—have recently shown promising results in handling sequential data. ViT model, in particular, has extended this capability to vision and time-series tasks by treating input sequences as image-like patches. Its self-attention mechanism makes it effective in learning long-term dependencies, something CNNs inherently struggle with. Yet, ViTs often underperform in capturing local, fine-grained details, especially in one-dimensional signals like ECGs.

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Our approach combines the best of both worlds. We propose ECG-ResViT, a hybrid architecture where CNNs are used for robust local feature extraction and ViTs for modeling long-range dependencies. To further enhance ViT performance, we integrate positional encoding to preserve the sequence of ECG signals, along with multi-head attention and feedforward layers for refined representation learning. This hybrid model is designed to strike a balance between accuracy, computational efficiency, and clinical interpretability.

While some approaches convert ECG signals into 2D formats like scalogram to fit vision-based models, such transformations may distort temporal information and add unnecessary complexity. Processing ECGs in their raw 1D form helps preserve time-dependent characteristics and keeps the model lightweight, which is crucial for real-time applications.

Although there has been extensive work with CNNs and a growing interest in Transformers, very few studies explore their hybridization for 1D biomedical signals like ECG. Moreover, existing solutions often overlook optimization strategies, which this study aims to address.

Beyond cardiovascular applications, signal analysis techniques, such as synchronization measures and network modeling, have shown immense value in neuroscience. For instance, recent studies using the phase-locking value (PLV) method have revealed that children with ADHD exhibit significantly higher neuronal synchrony and altered functional connectivity in the frontal and occipital lobes compared to healthy controls, indicating inefficient emotional processing pathways [2]. Complementing this, a broader review on neuronal synchrony patterns has emphasized the role of both pairwise and higher-order interactions in shaping cognitive dynamics, as well as their association with disorders like schizophrenia and Parkinson's disease [3]. These insights highlight the universal applicability of time-series analysis and synchronization modeling across both cardiac and cognitive domains, strengthening the relevance of robust signal-based modeling frameworks such as our proposed ECG-ResViT.

The remainder of this paper is organized as follows: Sect. 2 provides a comprehensive literature survey, discussing recent developments in ECG classification using DL, particularly CNNs, Transformers, and their hybrid variations. Section 3 outlines the proposed ECG-ResViT architecture, detailing its design, computational complexity analysis, and a discussion on its strengths and limitations. Section 4 presents the experimental setup and the results, including information on the dataset used, pre-processing techniques, evaluation metrics, ablation study, and a performance comparison with state-of-the-art methods. Finally, Sect. 5 concludes the study with key findings and potential directions for future research.

2 Literature survey

The automatic classification of ECG signals has gained significant attention due to its potential in detecting life-threatening cardiac anomalies. Traditional manual analysis of ECG signals is time-consuming and error-prone, leading to the need for reliable computer-aided diagnostic (CAD) systems. Various DL models, especially CNNs, have been extensively employed in this domain due to their ability to extract meaningful features directly from raw ECG signals.

Xu and Liu [4] proposed a CNN-based approach that achieved a remarkable overall accuracy of 99.43% for classifying arrhythmias using the MIT-BIH arrhythmia dataset. Similarly, Kachuee et al. [5] introduced a deep transferable representation for heartbeat classification, obtaining average accuracies of 93.4% and 95.9% for arrhythmia and myocardial infarction (MI) classification, respectively. Wang et al. [6] further enhanced CNN-based models by integrating Continuous Wavelet Transform (CWT) to capture time–frequency features, resulting in improved classification accuracy with an F1 score of 68.76% and accuracy of 98.74%.

Traditionally, CNNs have been highly effective in capturing local dependencies in ECG signals, identifying key patterns, such as P waves, QRS complexes, and T waves. However, CNNs often fall short when it comes to modeling long-term dependencies and global relationships in time-series data.

Ali et al. [7] conducted a comparative study between CNN, Support Vector Machine (SVM), and LSTM models, concluding that LSTM outperformed other methods due to its ability to capture temporal relationships and long-term dependencies. Singh and Mahapatra [8] proposed a recurrent CNN with Grey Wolf Optimization (GWO), enhancing classification accuracy to 98%. Similarly, Khan et al. [9] introduced a 1D convolutional deep residual neural network (ResNet) with SMOTE for addressing class imbalance, achieving an impressive average accuracy of 98.63%.

Mewada [10] explored a 2D-wavelet encoded CNN model to classify ECG signals as 2D images, which eliminated complex pre-processing steps like RR peak detection. The wavelet-based approach improved accuracy to 99.52%, indicating the potential benefits of multi-spectral feature integration. F. K R and Kurian [11] introduced C-BiNET, a hybrid model that combines CNNs and Bi-LSTM to capture both spatial and temporal dependencies. Their model achieved a classification accuracy of 98.24%, highlighting the advantages of incorporating bidirectional memory in ECG classification.

To address the challenges posed by multi-lead ECG signals, Yang et al. [12] proposed a multi-view and multi-scale deep neural network that considers different leads as different views, leveraging temporal features across multiple scales. Their approach achieved superior performance compared to existing networks. G. M. et al. [13] explored a DL framework for multi-label classification of ECG signals, addressing the challenge of identifying multiple heart conditions simultaneously. Their model incorporated explainable AI (XAI) using Grad-CAM to provide interpretability, achieving a subset accuracy of 96.2%.

Berrahou et al. [14] tackled the challenge of inter-patient ECG classification by combining entropy rate features with RR intervals in a CNN framework. Their model achieved superior accuracy of 99.13% and demonstrated robustness across datasets like MIT-BIH and INCART. Benchaira and Bitam [15] introduced a transfer learning approach using pre-trained Stacked CNN (SCNN) embeddings, which improved classifier performance across diverse datasets. This approach emphasized the importance of utilizing feature embeddings from diverse datasets to enhance ECG classification.

Daphin Lilda and Jayaparvathy [16] explored the classification of CVDs using spectrograms of 1D ECG signals, comparing the performance of 1D, 2D, and multi-branch CNN (MB-CNN) models. Their study demonstrated that MB-CNN achieved the highest classification accuracy of 99.34%, effectively capturing different levels of abstraction and improving classification performance.

The reviewed studies underscore the effectiveness of CNN-based approaches in ECG classification, with notable enhancements achieved by integrating wavelet-based methods, transfer learning, and multi-scale models. Hybrid models combining CNNs with LSTM and Bi-LSTM have proven effective in capturing both spatial and temporal dependencies, further improving classification accuracy. Additionally, multi-view, multi-scale, and multi-label models offer a promising avenue for addressing challenges in real-world ECG classification tasks, ensuring robust and interpretable results.

Recent studies have demonstrated the effectiveness of various DL models, including CNNs, Inception, ResNet, Recurrent Neural Networks (RNNs), LSTM, and Gated Recurrent Units (GRU), in classifying arrhythmias with remarkable accuracy. Pabitha et al. [17] conducted a comparative analysis of these models and found that ResNet achieved the highest accuracy of 98.4% in identifying user-specific ECG signals, highlighting the potential of ECG signals as unique biometric markers. Further advancements in ECG classification were explored by Daydulo et al. [18], who developed a DL model to classify ARR, CHF, and NSR using the MIT-BIH and BIDMC databases. Their approach achieved a remarkable 99.2% classification accuracy with superior sensitivity and specificity, demonstrating its potential to reduce diagnostic time and errors in ECG analysis.

Srinivas et al. [19] introduced an Improved Cuckoo Search Algorithm (ICSA) to optimize feature extraction in ECG classification. Using Discrete Wavelet Transformation (DWT) for noise removal and SVM with Feed Forward Back Propagation Neural Network (SVM-FFBPNN) for classification, the proposed ICSA model achieved 98.21% accuracy, outperforming competing algorithms, such as Improved Monarch Butterfly Optimization (IMBO) and Bat-Rider Optimization Algorithm (BROA). To address the challenge of limited patient-specific data, Malik et al. [20] proposed 1D Self-Organized Operational Neural Networks (1D Self-ONNs), which demonstrated superior performance over conventional CNNs in classifying supra-ventricular and ventricular ectopic beats. Their model achieved an average accuracy of 98% and 99.04% on the MIT-BIH dataset, showcasing its ability to learn adaptive operators without prior search, thus enhancing classification performance with minimal data. Pokaprakarn et al. [21] explored a novel DL architecture combining CNN and RNN layers to perform sequence-to-sequence cardiac rhythm classification. Their model effectively classified five types of cardiac rhythms using spectrograms and signal waveforms, achieving an average F1 score of 0.89 while maintaining a reduced number of training parameters.

In the realm of inter-patient ECG classification, Zahid et al. [22] proposed a compact 1D Self-ONN that leveraged both morphological and temporal features from heart cycles. Their approach achieved a remarkable 99.21% precision for normal (N) segments and 95.2% F1 score for ventricular ectopic beats (VEBs), outperforming state-of-the-art models while maintaining minimal computational complexity. Islam et al. [23] addressed the limitations of existing CNN and RNN models by introducing a hybrid model combining Bidirectional Gated Recurrent Units (BiGRU), Bi-LSTM, and multilayered dilated CNN. Their model, validated on the PhysioNet 2017 dataset, achieved 99.90% accuracy and demonstrated the ability to reduce computational time while enhancing classification performance.

Noise and baseline wander in ECG signals pose significant challenges in accurate classification. To mitigate this, Li et al. [24] proposed a Deep Score-Based Diffusion model for ECG (DeScoD-ECG) that effectively removed noise and baseline wander, achieving a 20% overall improvement in distance-based similarity metrics when compared with traditional filtering methods. Finally, Anitha et al. [25] introduced a Deep Bi-CapsNet architecture for precise feature extraction and classification of various arrhythmias using the MIT-BIH arrhythmia database. Their model, combining Deep Ensemble CNN-RNN with Bi-CapsNet, achieved 97.19% accuracy, outperforming traditional models, such as CNN, FTBO, and Capsule Network.

These recent advancements demonstrate the potential of DL techniques in enhancing ECG classification and arrhythmia detection. While CNNs remain powerful for capturing local patterns, the integration of advanced models, such as Self-ONNs, BiGRU-Bi-LSTM, and diffusion-based models, has significantly improved performance by addressing long-term dependencies, noise reduction, and inter-patient variability.

Recent advances in DL models have significantly improved the accuracy of ECG classification and CVD detection. ViT-based models, which have demonstrated exceptional performance in computer vision tasks, are now being explored for ECG classification. Hanoon Alkegany and Ali [26] introduced a deep ViT-based approach for ECG classification that integrates CNNs and self-attention mechanisms, achieving an impressive accuracy of 98.23% on benchmark ECG datasets. Similarly, Tao et al. [27] developed a novel DL model (2D ViT-1D CRNN) for heart sound signal classification by combining 1D CNN with Bi-LSTM for temporal feature extraction and using a 2D ViT module for time–frequency features. Their model outperformed state-of-the-art (SOTA) methods in two public datasets, underscoring its applicability for CVD diagnosis. Jamil and Roy [28] proposed an efficient and robust framework for valvular heart disease (VHD) classification using phonocardiogram (PCG) signals. Their approach combined 1D and 2D PCG features with nature-inspired optimization algorithms, such as particle swarm optimization (PSO) and genetic algorithms (GA), for feature selections. ViT applied to the time–frequency representation (TFR) of 2D PCG signals achieved a remarkable accuracy of 99.90% and an F1 score of 99.95%, outperforming existing VHD classification models. Telangore et al. [29] addressed the challenge of early prediction of sudden cardiac death (SCD) using a multimodal deep learning model that analyzed ECG signals at discrete intervals before SCD onset. Their approach utilized 1D CNN and LSTM for temporal pattern recognition, along with ViT and 2D CNN for feature extraction from scalograms and Hilbert spectra, achieving 100% classification accuracy.

Kim and Chung [30] tackled the challenges of time-series classification by introducing a ViT-based multi-scale classification framework that combines digital signal processing (DSP), image transformation, and ViT models. Their approach exhibited superior classification performance, particularly for complex and diverse time-series data. CHF diagnosis is another critical area where ViT has been effectively applied. Liu et al. [31] developed an ECG-Convolution-Vision Transformer Network (ECVT-Net) to detect CHF with an accuracy of 98.88% under an inter-patient scheme. The model's robustness against noise was validated through extensive experiments, making it suitable for real-world applications. Furthermore, Awan et al. [32] leveraged CNN-ViT-based ensemble classification for emotion recognition using physiological signals, including ECG and EEG. Their approach, validated on the AMIGOS and DEAP datasets, achieved outstanding performance with accuracies up to 98.2%, highlighting the potential of ViT-based models for dynamic and complex physiological data classification.

These studies collectively underscore the versatility and the efficiency of ViT-based models in addressing various challenges in ECG classification, SCD prediction, VHD detection, CHF diagnosis, PPG signal quality assessment, and emotion recognition. The integration of ViT with CNNs, LSTM, and advanced signal processing techniques has paved the way for more accurate and reliable automated healthcare systems. On the other hand, ViTs have emerged as powerful models capable of capturing global dependencies through self-attention mechanisms. While ViTs have demonstrated impressive results in analyzing 2D image data, their application to 1D ECG signals is still in its early stages.

Recent advances in DL models have significantly improved the accuracy of ECG classification and CVD detection. Hybrid CNN-Transformer models have emerged as promising solutions for time-series classification, particularly for ECG signals. Krishna et al. [33] introduced a Hybrid CNN-Transformer model with tuning techniques and GA optimization, achieving a remarkable 98.6% classification accuracy in ECG signal classification. Similarly, Liu et al. [34] developed a CNN-Transformer hybrid network for multi-class arrhythmia detection from PPG signals, demonstrating superior performance with an F1 score of 86.8% across multiple arrhythmia types. Li et al. [35] proposed a clinical knowledge-based ECG abnormality detection framework using dual-view CNN-Transformer and external attention mechanisms, achieving an average F1 score of 0.854 while enhancing interpretability and reducing complexity.

Lu et al. [36] explored the application of a CNN-Transformer model for classifying tetanus severity using ECG signals collected from low-cost wearable sensors. Their model achieved an accuracy of 88% with an AUC of 0.85, outperforming traditional methods. Anjum et al. [37] introduced a temporal Transformer-based fusion framework for arrhythmia classification that efficiently captured morphological changes in ECG signals, enhancing classification performance. Qiu et al. [38] applied a CWT with a multi-branch Transformer model for ECG classification, achieving 98.53% and 99.38% accuracy on the CPSC 2018 and MIT-BIH datasets, respectively. Their approach minimized pre-processing efforts and preserved critical time-series features. Kim et al. [39] proposed a hybrid CNN-Transformer model that utilized Stockwell transform for ECG feature extraction and outperformed SOTA models with 99.58% accuracy on the MIT-BIH dataset. Hou et al. [40] developed a Time-Hybrid OSA Transformer (THO) for obstructive sleep apnea (OSA) detection using single-lead ECG signals. Their model, which combined dilated convolution, LSTM, and a multi-scale feature fusion strategy, achieved 95.03% accuracy and an AUC of 96.85%, surpassing competitive models by 5%. Bentaleb and Khatar [41] introduced a multi-criteria Bayesian optimization framework for ECG signal denoising and classification using hybrid filters and empirical mode decomposition (EMD), improving ARR and MI classification accuracy to 99.61%. Finally, Islam et al. [42] presented CAT-Net, a convolution, attention, and Transformer-based model for single-lead ECG classification. CAT-Net demonstrated SOTA performance on both the MIT-BIH and INCART datasets, achieving a 99.14% accuracy and 94.69% macro F1 score for 5-class arrhythmia classification.

These studies collectively highlight the versatility and efficiency of hybrid CNN-Transformer and ViT-based models across diverse applications, including ECG classification, arrhythmia detection, VHD diagnosis, CHF prediction, OSA detection, and SCD prevention. The integration of convolutional, recurrent, and self-attention mechanisms with advanced signal processing techniques has led to significant improvements in classification accuracy, interpretability, and real-time performance, paving the way for more robust and automated healthcare systems.

Despite advancements in CNNs and Transformers, limited research has explored their hybridization for 1D ECG signal classification. Moreover, existing models often fail to strike a balance between computational efficiency and clinical interpretability, leaving a critical gap that needs to be addressed. Hybrid models that integrate CNNs and ViTs offer a promising solution by combining the best of both worlds—CNNs for extracting fine-grained local patterns and ViTs for modeling long-range dependencies. This synergy enhances classification performance by providing a comprehensive understanding of ECG signals.

3 Proposed methodology

This section introduces and elaborates on the proposed ECG-ResViT architecture, a novel approach designed to enhance classification performance by integrating the strengths of both CNN-based feature extraction and transformer-based temporal modeling. The rationale behind this design lies in addressing the limitations of conventional methods in capturing long-range dependencies and subtle morphological variations in ECG signals. A thorough analysis of the architecture, its advantages, and associated challenges is presented to highlight its practical significance and superiority over existing approaches. By combining residual connections and vision transformer blocks, the proposed model aims to improve both learning efficiency and classification accuracy, making it especially suitable for complex biomedical signal interpretation.

3.1 Proposed architecture

The architecture consists of a CNN feature extractor with residual blocks and dilated convolutions, followed by a ViT classifier with sinusoidal positional encodings and multi-head attention. The raw ECG signal is pre-processed by segmenting it into smaller segments and adding a channel dimension. The CNN extracts local and multi-scale features, which are then passed through a Global Average Pooling (GAP) layer to produce a fixed-size vector. This vector is reshaped and passed to the ViT, which models global dependencies in the ECG signal. The output of the ViT is aggregated using another GAP layer, and the final classification is performed using a softmax layer (Fig. 1).

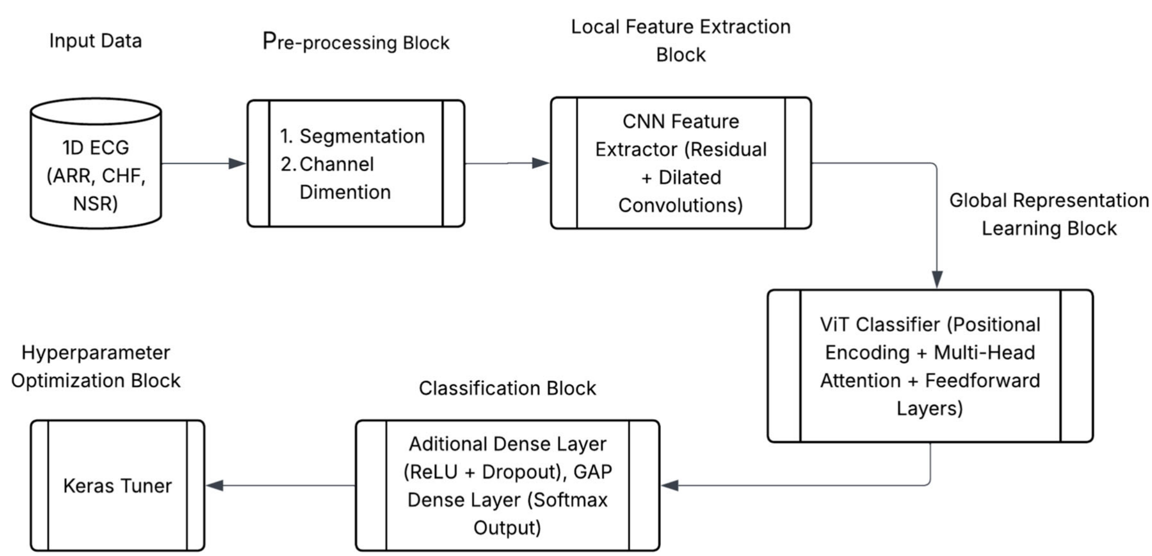

Fig. 1 Proposed ECG-ResViT architecture for ECG signal classification

3.1.1 Pre-processing block

The process starts with the raw ECG signal, a 1D time-series representation of the heart's electrical activity. However, these signals often come with challenges, they tend to be noisy and vary in length, making pre-processing a critical step before feeding the data into a DL model.

During pre-processing, two key transformations take place:

Segmentation: The ECG signal is divided into smaller, fixed-length segments (such as 500 samples per segment). This ensures that the input size remains consistent and helps the model focus on identifying local patterns within each segment.

Channel Dimension: To make the segmented signal compatible with Conv1D layers in a CNN, a channel dimension is added. This step is essential for preparing the data for convolutional operations, enabling the model to extract meaningful features effectively.

By performing these steps, the raw ECG signal is transformed into a structured format, ready for feature extraction, deep learning models to analyze and classify with greater accuracy.

3.1.2 Local feature extraction block

After pre-processing, the segmented ECG signals are passed through a CNN Feature Extractor to capture local and multi-scale patterns. The CNN architecture incorporates:

Residual Blocks: These blocks include skip connections that help mitigate the vanishing gradient problem and enable the training of deeper networks. Residual connections allow the model to learn identity mappings, making it easier to optimize.

Dilated Convolutions: Using different dilation rates (e.g., 2, 4, 8, 16, 32), the CNN can capture features at multiple temporal scales without significantly increasing the number of parameters. This is especially useful in ECG signals where patterns may occur across different time intervals.

The CNN extracts high-dimensional feature maps that encode rich information about the signal. However, to transition these features into a more compact and interpretable format, GAP is applied.

GAP reduces the high-dimensional feature map into a fixed-size vector by averaging across the time dimension. This step not only preserves important temporal information but also minimizes the risk of overfitting by summarizing the feature map into a compact representation. GAP ensures that the extracted features are in a format compatible with subsequent stages of the model, making them ready for classification or further processing.

By combining CNN Feature Extraction and GAP Aggregation, the model effectively learns hierarchical patterns in the ECG signals while retaining essential information in a reduced form, enhancing overall model efficiency and performance.

3.1.3 Global representation learning block

Traditionally, ViTs are designed to process 2D image data by dividing an image into patches and applying self-attention to model spatial relationships. In this work, ViT is adapted to process 1D ECG signals, leveraging self-attention mechanisms to analyze time-series data effectively.

Key Adaptations of ViT for 1D ECG Signals,

Sequence-to-Embedding Conversion: Instead of segmenting 2D images into patches, ECG signals are reshaped and projected into a sequence of embeddings, where each segment acts as an input token. This transformation allows the ViT to treat the ECG signal as a sequence of time steps, similar to how it processes image patches in traditional applications.

Positional Encodings for Temporal Context: To retain the order and timing of ECG events (such as P waves, QRS complexes, and T waves), sinusoidal positional encodings are added to the embeddings. This step is crucial when adapting ViTs to 1D signals, as the temporal context plays a significant role in interpreting ECG data.

Multi-Head Attention for Temporal Patterns: The multi-head attention mechanism in ViT processes different segments of the ECG signal simultaneously, capturing dependencies across time. This allows the model to learn complex temporal patterns and relationships that may not be apparent with convolutional layers alone.

3.1.4 Classification block

After processing the ECG signal through the ViT, the model proceeds to the classification block, which includes an additional dense layer with ReLU and dropout, a GAP layer, and a final output layer with softmax activation. These components play a crucial role in transforming the sequence of embeddings from the ViT into the final classification decision.

Additional Dense Layer (ReLU + Dropout): The sequence of embeddings produced by the ViT is first passed through a fully connected dense layer with ReLU activation. This layer introduces non-linearity and allows the model to learn complex feature interactions. A dropout mechanism is also applied here to prevent overfitting by randomly deactivating a fraction of neurons during training.

GAP Layer: The output of the dense layer is then passed through the GAP layer, which performs the following key functions:

1. **Aggregation of Temporal Information:** The GAP layer averages across the time dimension of the processed embeddings, reducing them into a single fixed-size vector. This operation summarizes the temporal information present in the ECG signal.
2. **Dimensionality Reduction:** By transforming a variable-length sequence into a fixed-size representation, it simplifies the structure for the final layer and reduces the risk of overfitting.
3. **Robustness to Variable-Length Inputs:** Since GAP averages over time, it produces a consistent output size regardless of the input length, enabling flexibility in processing varying ECG signal lengths.

The result of the GAP layer is a compact feature vector that encapsulates the most informative aspects of the ECG signal for classification.

Final Output Layer (Dense + Softmax): This fixed-size vector is passed through a fully connected (dense) output layer followed by a softmax activation:

1. **Fully Connected Layer**: Maps the compact representation to a set of logits, each corresponding to a possible class label (e.g., ARR, CHF, NSR).
2. **Softmax Activation:** The softmax function converts the logits into probabilities, ensuring that the output is interpretable and suitable for classification. Mathematically softmax function can be defined by (1),

$$Softmax(Z_i) = \frac{e^{Z_i}}{\sum_{j=1}^{C} e^{Z_j}}, \tag{1}$$

where Z_i = logit for Class i, C = Total number of Classes (C = 3).

Final Classification: The class with the highest probability is selected as the model's prediction. This provides a clear and interpretable output for clinical decision-making.

3.2 Computational complexity analysis

The computational complexity of the proposed ECG-ResViT model can be analyzed based on two major processes:

(i) the CNN-based feature extraction, and
(ii) the Transformer-based classification head.

3.2.1 CNN feature extraction

The input to the model is a 1D ECG signal of length $n = 500$ with 1 channel. The CNN feature extractor consists of multiple convolutional layers including dilated convolutions and residual connections. Let: k be the kernel size, d be the number of input channels (here, $d = 1$), f be the number of filters per layer, and L be the number of convolutional layers.

The computational complexity of a single 1D convolution operation is $O(nkdf)$ [43], and for L such layers, the total complexity becomes: $O(nkdf)$

From our model summary, the CNN feature extractor has approximately 1.75 million parameters, which indicates a moderate depth and width of convolutions. GAP follows, which has negligible computational cost compared to convolutions. The final output of the CNN is a 512-dimentional feature vector.

3.2.2 ViT classification block

This 512-dimensional output is passed through a dense layer to generate a 640-dimensional vector, which is reshaped into a sequence of 10 tokens of dimension 64 — i.e., $T = 10$, $N = 64$.

In the Vision Transformer, the complexity of the self-attention mechanism per layer is [43]: $O(T^2N + TN^2)$. This is relatively lightweight and feasible for real-time inference on 1D signals. The ViT block includes: Multi-head self-attention, Feedforward dense layers and Layer normalization and positional encoding. After processing, a GAP layer is again applied across the token dimension, followed by a final dense classification layer with 3 output classes, having negligible cost $O(NC) = O(64 * 3) = O(192)$

3.2.3 Overall complexity

Combining both stages, the total computational complexity of the model is approximately: $O(nkdfL) + O(T^2N + TN^2) + O(NC)$. Since k, d, f, L < < N and T < < N, the dominant term becomes, indicating that the self-attention mechanism governs the model's complexity as the input grows.

3.2.4 Efficiency insight

Despite its hybrid structure, the model maintains computational efficiency due to: use of GAP layers instead of fully connected ones early on, a compact tokenization strategy (only 10 tokens), Lightweight ViT design with Linformer-style reduction, and Operating entirely on raw 1D signals, avoiding costly 2D transformations. This makes ECG-ResViT not only accurate but also efficient and scalable for real-time ECG classification tasks.

3.3 Advantages of ECG-ResViT architecture

ECG-ResViT architecture will give better metrics because of the following reasons,

1. The CNN captures local patterns (e.g., P waves, QRS complexes) through hierarchical convolutions, while the ViT models' long-range temporal dependencies using self-attention. This combination enables the model to learn both fine-grained and global features, improving classification accuracy for complex ECG patterns.
2. The use of GAP after both the CNN and ViT ensures compatibility with variable-length ECG signals, making the architecture suitable for real-world data where signal duration varies.
3. The ECG-ResViT architecture achieves high accuracy by leveraging the strengths of CNNs and ViTs. Additionally, the ViT's self-attention mechanism provides interpretability by highlighting important temporal segments of the ECG signal, aiding clinical decision-making.

3.4 Limitations of ECG-ResViT architecture

Even though ECG-ResViT architecture gives better metrics, it has some limitations:

1. The computational and memory demands of the ViT make it difficult to deploy the model in real-time applications (e.g., wearable devices) without significant optimization or hardware acceleration.
2. The ViT's self-attention mechanism has a quadratic complexity $\mathrm{O(T^2N)}$, making it computationally expensive for long sequences. Additionally, storing $\mathrm{T} \times \mathrm{T}$ attention matrices consumes significant memory, limiting deployment in resource-constrained environments.

4 Experimental results and discussions

This section provides a comprehensive overview of the dataset, outlines the implementation details, defines the metrics utilized for evaluation, and presents the experimental results along with related discussions.

4.1 Implementation details

This study employs ECG data gathered from three separate classes: individuals diagnosed with ARR, those experiencing CHF, and individuals exhibiting NSR. The dataset comprises 162 ECG recordings, obtained from three reputable PhysioNet databases: the MIT-BIH Arrhythmia Database [44, 45], the MIT-BIH Normal Sinus Rhythm Database [45], and the BIDMC Congestive Heart Failure Database [46]. A total of 96 recordings are associated with individuals experiencing ARR, 30 recordings pertain to those diagnosed with CHF, and 36 recordings illustrate individuals exhibiting NSR.

For a comprehensive evaluation, we choose 30 recordings from each category for additional processing. The recordings are subjected to a segmentation process, dividing each ECG signal into 10 distinct, non-overlapping segments, with each segment comprising 500 data points. This step improves the organization of the dataset, enabling the model to identify significant patterns within the signals while maintaining adequate data for both training and evaluation purposes.

Through the systematic organization and pre-processing of the data, our objective is to create a reliable classifier that can effectively differentiate among the three cardiac conditions. This systematic method guarantees that the model is provided with a balanced and representative dataset, leading to more dependable classification results. The dataset for our work is referenced from https://in.mathworks.com/help/deeplearning/ug/classify-time-series-using-wavelet-analysis-and-deep-learning.html

4.2 Evaluation metrics

The following metrics were used to evaluate the performance of the proposed ECG-ResViT architecture:

Accuracy: Accuracy measures the proportion of correctly classified instances out of the total instances. It is calculated as (2)

$$Accuracy = \frac{TruePositive(TP) + TrueNegative(TN)}{TP + TN + FalsePositive(FP) + FalseNegative(FN)}. \tag{2}$$

A high accuracy (in our case its 99.13%) indicates that the model correctly classifies most ECG signals.

Precision: Precision quantifies the proportion of TP predictions among all positive predictions. It is defined as (3)

$$Precision = \frac{TP}{TP + FP}. \tag{3}$$

High precision (in our case its 99.15%) ensures that the model minimizes FP, making its positive predictions highly reliable.

Recall (Sensitivity): Recall measures the proportion of true positives identified out of all actual positives. It is computed as (4)

$$Recall = \frac{TP}{TP + FN}. \tag{4}$$

A high recall (in our case its 99.13%) indicates that the model effectively identifies most positive cases, reducing the risk of missed diagnoses.

F1 Score: The F1 score is the harmonic mean of precision and recall, providing a balanced measure of the model's performance. It is calculated as (5)

$$F1Score = 2 * \frac{Precision * Recall}{Precision + Recall}. \tag{5}$$

An F1 score close to 1.0 (in our case its 99.13%) indicates a robust balance between precision and recall.

ROC-AUC: The Receiver Operating Characteristic (ROC) Area Under the Curve (AUC) measures the model's ability to distinguish between classes. It is the area under the plot of the True Positive Rate (TPR) against the False Positive Rate (FPR). An AUC close to 1.0 (in our case its 0.9994) signifies excellent discriminative power.

4.3 Ablation study

We conducted a detailed ablation study to analyze how each architectural and training enhancement contributes to the overall performance of our ECG classification model. The progression across the three models is designed to isolate and evaluate the impact of convolutional design, attention mechanisms, regularization techniques, and hyperparameter tuning. Performance metrics, such as accuracy, precision, recall, and F1 score, were used for comparison are described in Table 1.

To understand the impact of successive architectural and training enhancements, we conducted an ablation study across three progressively advanced models. The baseline model (Base-CNN-ViT) combined a standard convolutional neural network (CNN) with the AdamW optimizer and a Vision Transformer (ViT) employing Linformer attention, applied directly on 1D ECG signals. Despite its lightweight configuration, comprising approximately

Table 1 Ablation study on different architectures for ECG classification

S. no	Architecture	Parameters	Accuracy	Precision	Recall	F1 score
1	Base-CNN-ViT	973,123	96.30	96.40	96.30	96.29
2	ResDilate-CNN-ViT + LS	2,203,971	97.78	97.92	97.78	97.79
3	Tuned-ResDilate-CNN-ViT + LS	6,611,915	99.13	99.15	99.13	99.13

Base-CNN-ViT Basic CNN + ViT with Linformer, *ResDilate-CNN-ViT + LS* CNN with dilated convolutions, residuals + ViT + label smoothing, *Tuned* includes hyperparameter tuning

973 K parameters, this model achieved strong performance with an accuracy of 96.30%, precision of 96.40%, recall of 96.30%, and an F1 score of 96.29%, thereby validating the efficacy of a CNN-Transformer hybrid for ECG classification tasks.

Building upon this foundation, ResDilate-CNN-ViT + LS introduced architectural improvements by incorporating dilated convolutions and residual connections into the CNN feature extractor. These modifications enabled the network to better capture long-range temporal dependencies and enhance feature reuse. Additionally, label smoothing was employed during training to promote generalization and mitigate overfitting. This enhanced architecture, with approximately 2.2 million parameters, delivered improved results—achieving 97.78% accuracy, 97.92% precision, 97.78% recall, and 97.79% F1 score—highlighting the value of deeper receptive fields and regularization strategies.

Tuned-ResDilate-CNN-ViT + LS further refined the previous design by introducing hyperparameter optimization using Keras Tuner. This tuning process explored a search space of learning rates, batch sizes, dropout rates, and training epochs, ultimately identifying an optimal configuration: a learning rate of 0.001, weight decay of 1e-6, dropout rate of 0.3, batch size of 16, and 40 training epochs. As a result, the model's parameter count increased to approximately 6.6 million, but the classification performance significantly improved—reaching 99.13% accuracy, 99.15% precision, 99.13% recall, and 99.13% F1 score.

Collectively, this ablation study demonstrates that each stage—from the baseline hybrid architecture to the integration of advanced convolutional strategies and hyperparameter tuning—yields measurable performance gains. The final model not only showcases superior accuracy but also underscores the crucial role of both architectural innovation and training optimization in biomedical signal classification.

For statistical validation, we performed a one-way ANOVA on the accuracy values of the last three models like Base-CNN-ViT, ResDilate-CNN-ViT + LS and Tuned-ResDilate-CNN-ViT + LS as reported in Table 1 based on fivefold cross-validation results.

While ANOVA typically assumes the data follows a normal distribution, we proceeded with the analysis using the raw accuracy values produced by our DL models to maintain practical relevance. We recognize that these values may not perfectly meet the assumption of normality, but they still offer useful insight into the variation in model performance.

The test produced an F statistics of 2.0493. When compared with standard F critical values, the F statistics did exceed the critical value at the 80% confidence level (F critical = 1.8460) which indicates that there are some meaningful differences among the performances of models reported in Table 1. However, the result was not statistically significant at common confidence levels, such as 95% (F critical = 3.8853), 97.5% (F critical = 5.0959), or even 90% (F critical = 2.8068). It can be inferred that, owing to the fact that ANOVA requires samples to follow normal distribution which is actually not met by the accuracy values resulting from five-fold cross validation, the test confirms the significant difference only at 80% confidence level and not at stricter levels. It is also worth acknowledging the fact that the existing models did produce significant accuracy values, say 96% or higher, and our proposed model aims at improving them.

4.4 Performance comparison with the state-of-the-art approaches

Table 2 gives a clear picture of how different deep learning models have performed in classifying ECG signals into three key categories, ARR, CHF, and NSR. Each study listed in the table brings a unique approach to tackling this problem. While some have relied on traditional CNN architectures for feature extraction, others have explored hybrid models or transformed ECG signals into 2D spectrograms before classification.

We see models like AlexNet [48], ResNet18 [49], and SqueezeNet [50] used in various forms —either alone, with transfer learning, or combined with signal transformation techniques like CWT [50] or CQ-NSGT [51]. Notably, the proposed CNN-ViT model, which blends the strengths of CNN and ViT, stands out with the highest reported accuracy and AUC score. This is likely due to its ability to capture both fine-grained local features and long-term global dependencies—something that's particularly important for analyzing ECG signals.

Table 2 SOTA performance of deep learning models on ECG classification (ARR, CHF, NSR)

S. no	Study/paper	Methodology	Model used	Remarks	Accuracy/metrics
1	Khadirnaikar et al. (2020) [47]	1D + Hybrid AlexNet + SVM	AlexNet for feature extraction, SVM for classification	Outperforms traditional ML classifiers like SVM (68.75%) & KNN (65.63%)	96.77% (Acc)
2	Sabeenian et al. (2023) [48]	Transfer Learning (TL) on CNNs	ResNet18 & DenseNet	Used pre-trained CNNs with fine-tuning for ECG classification	ResNet18: Raw ECG: 98.81% Wavelet data: 97.05%
3	Nitish Katal et al. (2023) [49]	CNN vs GoogLeNet	CNN & GoogLeNet	Smaller CNN outperformed deeper GoogLeNet in 1D ECG signal setting	CNN: 91.20%
4	Sankar Ganesh et al. (2024) [50]	CWT + 2D Spectrograms + SqueezeNet	SqueezeNet CNN	Achieved highest performance using image-based transformation	98.7% (Acc)
5	Manjur Kolhar et al. (2024) [51]	combines the Constant Q Non-Stationary Gabor Transform (CQ-NSGT) to convert 1D ECG signals into 2D time–frequency representations	CQ-NSGT + AlexNet (Time–Frequency + CNN)	Achieved high validation and test accuracy with proper data pre-processing	98.82% (Acc)
6	**Proposed**	**Hybrid CNN-ViT with Residual & Dilated Convolutions, Positional Encoding & Multi Headed Attention**	**CNN (local features) + ViT (global dependencies)**	**Combines spatial and temporal learning in 1D signal; interpretable and efficient**	**99.13% (Acc) 99.94% (ROC—AUC)**

Overall, the table not only highlights how far ECG classification models have come but also emphasizes the value of combining different techniques to improve reliability, speed, and real-world applicability—especially when used in clinical environments.

5 Conclusion

In this study, we introduced ECG-ResViT, a hybrid deep learning architecture that effectively combines the strengths of CNNs and ViTs for classifying ECG signals into ARR, CHF, and NSR categories. By leveraging residual and dilated convolutions, the CNN component captures rich local patterns, while the ViT component, enhanced with positional encoding and multi-head attention, models long-range dependencies crucial for understanding temporal dynamics in ECG data. Our approach demonstrates superior performance compared to existing state-of-the-art models, both in terms of accuracy and interpretability. The results highlight the potential of integrating CNNs and ViTs in medical signal processing, especially in real-time diagnostic settings. Future work will focus on further optimizing the model for real-time deployment, expanding to multi-class ECG datasets, and enhancing explainability to support clinical decision-making.

Future work will focus on validating the ECG-ResViT model across larger, diverse ECG datasets. We aim to enhance computational efficiency for real-time and edge device deployment. Integration of explainable AI (XAI) will improve interpretability for clinical use. Patient-specific adaptive learning can be explored to personalize ECG classification.

Acknowledgements The authors express their gratitude to Vellore Institute of Technology (VIT), Chennai, India, for providing valuable support to carry out this work at VIT.

Author contributions

The study was conceived and designed by Premanand S and Sathiya Narayanan. Data collection was performed by Premanand S. The analysis and interpretation of results were done by Premanand S and Sathiya Narayanan. Premanand S prepared the initial draft of the publication. The results were evaluated by all authors and the final version of the publication was approved by them.

Funding The authors received no specific funding for this study.

Data availability The datasets analyzed during the current study are available in the following link: https://in.mathworks.com/help/deeplearning/ug/classify-time-series-using-wavelet-analysis-and-deep-learning.html

Code availability Programming codes and the data supporting the findings of this study are available upon reasonable request.

Declarations

Conflict of interest The authors declare that there are no conflicts of interest regarding the publication of this paper.

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